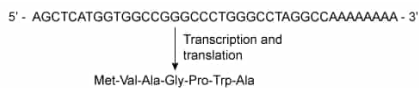


A protein that results from transcription and translation of a gene is shown.



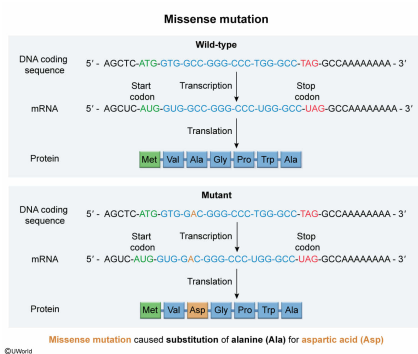
A mutation in this gene results in the replacement of the alanine at the third amino acid position with aspartic acid. This type of DNA mutation would most likely be defined as:

- ☐ A. a frameshift mutation. [4%]
- ☐ B. a nonsense mutation. [5%]
- ✓ ☒ C. a missense mutation. [85%]
- ☐ D. a silent mutation. [4%]

Correct

86% answered correctly

Explanation:



Exposure to mutagens (manmade or environmental) or **errors** in **DNA replication** may cause **mutations** (ie, changes) in DNA sequences. **Transcription** and **translation** of mutant DNA leads to the production of mutant (ie, altered) proteins. There are several types of DNA mutations, including:

- **Frameshift mutations**, which result from the insertion or deletion of a number of nucleotide base pairs *not* divisible by three. This causes the **reading frame** of the mRNA transcript to change, often leading to *multiple* amino acid changes downstream of the mutation, which can potentially interfere with protein function (**Choice A**).
- **Nonsense mutations**, which occur when a base pair substitution causes a premature **stop codon** to be inserted into a gene. This premature stop codon leads to early termination of translation and the production of a truncated (ie, shortened) protein (**Choice B**).
- **Missense mutations**, which occur when a base pair substitution results in a codon that encodes a **different amino acid**. This results in the **substitution** of the wild-type amino acid in the protein for this different, **mutant** amino acid, potentially interfering with protein function. However, there is a possibility that protein function may not be affected if an amino acid with a similar structure (ie, properties) is substituted for the original amino acid.
- **Silent mutations**, which occur when a mutation alters the sequence of the gene and mRNA transcript but *not* the sequence of amino acids in the encoded protein. As a result, the protein retains its wild-type function and is *not* affected by silent mutations (**Choice D**).

The question diagram shows the transcription and translation of a gene to produce a protein. In this scenario, a mutation occurring in this gene results in the replacement of the alanine at the third amino acid position with aspartic acid. Because this mutation caused the **substitution** of **one amino acid for another**, it is most likely a **missense mutation**.

Educational objective:

DNA mutations may be caused by DNA replication errors or exposure to mutagens. Missense mutations occur when a base pair substitution in a gene causes one amino acid to be substituted for another in the encoded protein.

Time spent: 0 sec
Subject:
Biology

QID: 403124
System:
Molecular Biology